

1

Bottle

(OBB-aligned tall-cylinder grasp)



Problem

Wine bottles tip during grasp when the gripper closes with arbitrary yaw — the grasp contacts the cylinder at a glancing angle and the bottle rolls out.



When to Apply

- Tall cylindrical objects: wine, tall cans, spray bottles
- Aspect ratio > 2.0



Strategy

Align gripper yaw to the OBB major axis so jaws close across the cylinder diameter. Use a two-stage close: scout at 50%, seat at 70%, then slow lift.



Bottle

```

obb = compute_obb(bottle_points)
major = obb.axes[0]
grasp_yaw = np.arctan2(major[1], major[0])

goto_pose(bottle.centroid, yaw=grasp_yaw)
set_gripper_width(0.5 * MAX_WIDTH)    # scout
set_gripper_width(0.7 * MAX_WIDTH)    # seat
lift_slowly(dz=0.15, speed=0.05)

```

2

Box

(small-flat centroid grasp)

3

Air-Grasp Detection

(post-close verification)

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Strategic Grasping